THE DIAGONAL OF AN AFFINE FORMAL ALGEBRAIC SPACE IS REPRESENTABLE AND A CLOSED IMMERSION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Affine formal algebraic spaces as filtered colimits, with Zariski local factorisation of a morphism through a stage. Same chapter, Definition definition-affine-formal-algebraic-space and the discussion following it, invoked at proof lines 2019-2024 and 2030-2034. Supplies the factorisation used in both parts.
- A representable diagonal, and the diagonal of a separated morphism of schemes is a closed immersion. Spaces, Lemma spaces-lemma-representable-diagonal, invoked at proof lines 2024-2026. Converts the scheme-wise fibre product computation into representability of the diagonal, and identifies the diagonal of an affine scheme with a closed immersion.

Source reference. Original lemma 025W: sources/stacks-chow/tags.txt, source lines 2829–2829. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

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such that

$$X \cong \operatorname{colim}_{\lambda \in \Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. If X is an affine formal algebraic space over S, then the diagonal morphism $\Delta : X \to X \times_S X$ is representable and a closed immersion.*

Proof. Suppose given $U \to X$ and $V \to X$ where U, V are schemes over S. Let us show that $U \times_X V$ is representable. Write $X = \operatorname{colim} X_\lambda$ as in Definition 0.1. The discussion above shows that Zariski locally on U and V the morphisms factors through some X_λ. In this case $U \times_X V = U \times_{X_\lambda} V$ which is a scheme. Thus the diagonal is representable, see Spaces, Lemma 025W. Given $(a, b) : W \to X \times_S X$ where W is a scheme over S consider the map $X \times_{\Delta, X \times_S X, (a,b)} W \to W$. As before locally on W the morphisms a and b map into the affine scheme X_λ for some λ and then we get the morphism $X_\lambda \times_{\Delta_\lambda, X_\lambda \times_S X_\lambda, (a,b)} W \to W$. This is the base change of $\Delta_\lambda : X_\lambda \to X_\lambda \times_S X_\lambda$ which is a closed immersion as $X_\lambda \to S$ is separated (because X_λ is affine). Thus $X \to X \times_S X$ is a closed immersion. $\square$